

Swarup G L

Phone: +91 7676603557
Email: swarupthippyswamy@gmail.com
Address: Holalkere(T), Chitradurga(D), Karnataka, 577526
GitHub: github.com/Swarupgl



Professional Summary

Motivated and innovative B.Tech student in Artificial Intelligence and Data Science at IIIT Raichur. Dedicated to applying knowledge in AI, data science, IoT, and web development to solve real-world problems and develop impactful projects. Seeking opportunities to expand skills and contribute to advanced technology development.

Education

Indian Institute of Information Technology (IIIT), Raichur	Raichur, Karnataka
B.Tech in Artificial Intelligence and Data Science	2023 – 2027

Skills

Programming Languages	JavaScript, Python, C, C++, Dart, HTML, CSS.
Web Development	React.js, Next.js(basic), Node.js, Express.js, Tailwind CSS, REST APIs
AI & Machine Learning	TensorFlow, 6G ISAC sensor fusion, Multi-modal object detection
Mobile & Edge IoT	Flutter, Android development(basic), BLE communication, ESP32, Arduino IDE
Backend Architectures	Scalable RESTful design, Secure decentralized file handling, SQL, mongo db
Version Control	Git, GitHub (for standard code collaboration)

Work Experience

Frontend Developer Intern(1 month)	<i>InternLoom</i>	Remote
- Developed responsive frontend user interfaces for an internship management platform by accurately translating Figma design specifications into functional code. - Designed and updated interactive UI components, including dynamic data cards, to effectively render and display user information within the application. - Collaborated with the engineering team using Git version control workflows to integrate frontend features and ensure seamless code deployment.		

Technical Coordinator

ElectroGeeks Club, IIIT Raichur

2024 – 2026

Raichur, India

- Coordinated multiple hands-on hardware workshops on ESP32, component layout structures, and technical architecture sessions for stable BLE communication and multimodal data routing.
- Mentored and guided multi-disciplinary student teams to plan, structure, and successfully execute multi-sensor fusion logic and physical alerts triggers for end-to-end engineered system exhibitions.

Ghost-Eye — 6G Integrated Sensing and Communication (ISAC) System(ongoing)

Thermal Sensor Fusion, MATLAB Debugging, MATLAB Multi-modal Object Detection, Python

- Engineered a custom object detection system tailored for vehicular safety in low-visibility environments, integrating multi-modal data streams from radar, thermal cameras, and standard image feeds.
- Focused research on optimizing communication path loss algorithms to account for multi-path fading in dense 6G network simulations.
- Translated raw sensor outputs into actionable situational awareness, demonstrating systems architecture mastery with complex non-traditional data pipelines.

Verdict Vault — Decentralized Secure Legal Ledger

Node.js, Express.js, Blockchain, IPFS, MongoDB

- Architected a decentralized document management system designed to coordinate high-security document interactions between multiple legal stakeholders.
- Merged decentralized file storage logic (IPFS) with a custom standard backend API to ensure unforgeable cryptographic ledger integrity and optimized data access controls.

Leaderboard(contribution in building college application)

JavaScript, Node.js, Express.js, MongoDB

- Developing a web-based platform to track and display running progress and rankings in real time.
- Responsible for backend development and API design, ensuring robust and efficient database interactions and integrations.

Dormant Device Detection and Tracking System

ESP8266, MOSFETs, Voltage Boosters, GPS Modules, Mobile App Integration

- Designed an innovative low-power circuit configuration using a dynamic 10% threshold power-switchover mechanism to prolong battery life of tracking sensors significantly.
- Interfaced raw GPS coordinates via optimized hardware serial parsing directly to a mobile application UI for live mapping visualisation.

Bio Bridge(Smart Attendance System) — Cross-Platform IoT Bridge

Flutter, Dart, BLE Communication, ESP32, SQLite

- Engineered a direct cross-platform mobile bridge establishing stable data flows between physical ESP32 biometric scanners and dynamic tables on SQLite/Google Sheets.
- Structured automated, resilient REST API pipelines to execute secure data pushes based on transactional success codes, ensuring data flow continuity in challenging connection scenarios.

Declaration

I hereby declare that the details and information given above are complete and true to the best of my knowledge.

Swarup G L